

Model Question Paper – SET 1

ANTENNA AND WAVE PROPAGATION

Time: 3 Hour

Maximum Marks: 75

PART A

- I. Answer **all** questions in one word or one sentence. Each question carries 1 mark
(9*1=9 marks)

Module outcome Cognitive level

1	List the modes of radio wave propagation.	M 1.01	R
2	Define duct propagation.	M 1.03	R
3	Define critical frequency.	M 2.02	R
4	What is the relation between MUF and skip distance?	M 2.02	R
5	What is fading?	M 2.03	R
6	Define half power bandwidth.	M3.03	R
7	Gain of antenna means?	M 3.03	R
8	List any two applications of micro strip antenna.	M 4.02	R
9	What are the disadvantages of Yagi-Uda Antenna?	M 4.03	R

PART B

- II. Answer any **eight** questions from the following, each question carries 3 marks.
(8 * 3=24 marks)

Module outcome Cognitive level

1	Explain the mechanism of tropospheric propagation.	M 1.03	U
2	Define maximum usable frequency.	M 2.02	R
3	What are the disadvantages of ground wave propagation?	M1.02	R
4	What are the effects of obstacles in radio wave	M 2.03	R

	propagation?		
5	Outline the importance of radiation resistance of an antenna.	M 3.03	U
6	Define duality of antenna.	M 3.04	R
7	What does radiation pattern of an antenna means?	M 3.03	R
8	Write notes on loop antenna.	M4.01	U
9	List the merits and demerits of aperture antennas.	M4.02	R
10	Write notes on helical wire antenna.	M4.03	U

PART C

III. Answer ALL questions. Each question carries 7 marks. (6*7=42 marks)

Module outcome Cognitive level

1	Outline the effects of earth's curvature on space wave propagation?	M 1.03	U
	OR		
2	Summarize the factors affecting the field strength of ground wave propagation?	M 1.02	U
3	Explain the structure of ionosphere.	M 2.01	U
	OR		
4	What is diversity? What are the different diversity techniques?	M 2.03	U
5	Define the terms a) virtual height b) skip distance c) ray path	M2.02	R
	OR		
6	Explain the reflection and refraction of sky waves by ionosphere.	M 2.01	U

7	Describe the radiation mechanism of an antenna.	M 3.01	U
	OR		
8	Define the following parameters a) radiation resistance b) directivity c) signal to noise ratio	M3.03	R
9	Explain horn antenna.	M 4.02	U
	OR		
10	Explain Yagi-Uda antenna.	M 4.03	U
11	Write notes on reflector antenna.	M 4.03	U
	OR		
12	Explain different types of antenna arrays?	M 4.04	U

Model Question Paper – Set 2

ANTENNA AND WAVE PROPAGATION

Time: 3 Hour

Maximum Marks: 75

PART A

I. Answer **all** questions in one word or one sentence. Each question carries 1 mark.

Module outcome Cognitive level

1	Define wave tilt.	M 1.02	R
2	What is line of sight propagation?	M 1.02	R
3	What do you mean by skip distance?	M 2.02	R
4	Define virtual height.	M 2.02	R
5	Define multihop propagation.	M 2.03	R
6	Define directivity.	M 3.03	R
7	What is antenna temperature?	M 3.03	R
8	What are the advantages of antenna arrays?	M 4.04	R
9	What is dipole?	M 4.01	R

PART B

II. Answer any **eight** questions from the following, each question carries 3 marks.
(8 * 3=24 marks)

Module outcome Cognitive level

II.1	What are the different modes of radio wave propagation?	M 1.01	R
II.2	What are the advantages of ground wave propagation?	M 1.02	R
II.3	What is the relation between MUF, critical frequency and skip distance?	M 2.02	U
II.4	Outline the effects of ionosphere absorption?	M 2.01	U
II.5	Explain sky wave propagation.	M 2.02	U
II.6	Define effective aperture of an antenna.	M 3.03	R
II.7	Define reciprocity of an antenna.	M 3.04	R

II.8	Differentiate the terms beam area and beam width of an antenna.	M 3.03	U
II.9	Write notes on helical antenna.	M 4.03	U
II.10	What is a half wave dipole antenna?	M 4.01	R

PART C

Answer ALL questions. Each question carries 7 marks. (6*7=42 marks)

Module outcome Cognitive level

III.1	Outline the atmospheric effects on space wave propagation?	M 1.03	U
	OR		
III.2	Explain about Ground wave propagation.	M 1.02	U
III.3	Explain how field strength of space wave propagation is affected by distance and height.	M 1.03	U
	OR		
III.4	Explain duct propagation.	M 1.03	U
III.5	Define fading and the different types of fading.	M 2.03	R
	OR		
III.6	Define the terms a) diversity b) critical frequency c) skip zone	M 2.02	R
III.7	Explain radiation pattern and effective aperture of an antenna.	M 3.03	U
	OR		
III.8	What is antenna? What are the different functions of antenna?	M 3.02	R
III.9	Explain the working of loop antenna. What are its application?	M 4.01	U
	OR		
III.10	Explain the working of linear antenna array and its significance.	M 4.04	U
III.11	Explain the working of slot antenna.	M 4.02	U
	OR		
III.12	Explain the working of reflector antenna.	M 4.03	U